



A history lesson!?

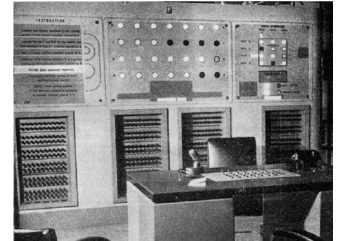


- Why should we care?
- There were certain events, design concepts, influential persons and technological advancements which had major impact on how games are being developed today!
 - Players and developers alike take a lot of things for granted
 - Graphics, color, sound, ergonomically designed controllers, savable game state, exchangeable game disks, reward systems, perceivable feedback, etc.
 - You can learn a lot from the mistakes others made
 - It's good to know what has been done before
- We will have a quick look at some of the most important events on the video game timeline, of which every game developer should know about.

1947 - 1952



- 1947: Cathode Ray Tube Amusement Device
 - The earliest known interactive electronic game was by Thomas T. Goldsmith Jr. and Estle Ray Mann on a cathode ray tube.
- 1947: Chess
 - Alan Turing developed a theoretical computer chess program as an example of machine intelligence.
- 1950: Imitation game
 - Proposed (only theoretically) by Alan Turing in his article “Computing Machinery and Intelligence,” published in Mind magazine.
- 1951: Nim
 - The NIMROD computer, created by Ferranti, was presented at the Festival of Britain.
- 1952: OXO
 - Alexander S. Douglas made the first computer game to use a digital graphical display.



NIMROD (1951)

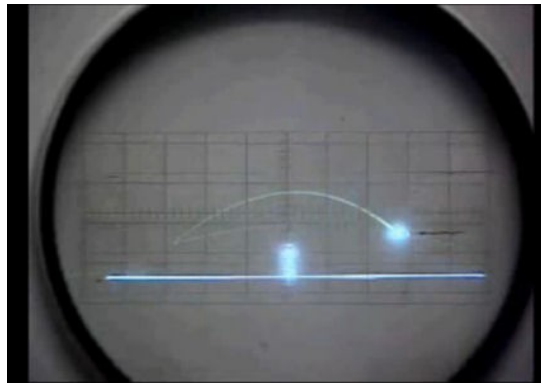


OXO (1952)

1958: Tennis for Two



- Willy Higinbotham at Brookhaven National Lab designed it for an analog computer.
- This game, played on an oscilloscope display, was significant in that it was the first game to permit two players actually to control the direction and motion of the object moving on the screen (the ball).
- The object of the game was to maintain a volley for as long as possible by hitting the ball with one of the two rackets at each side of the screen. A line down the middle indicated the net; gravity, bounce, and wind speed were calculated into game play.



<http://www.youtube.com/watch?v=u6mu5B-YZU>

8

1959 - 1961



- 1959: Mouse in the Maze, Tic-Tac-Toe
 - A collection of interactive graphical programs were created on the TX-0 experimental computer at MIT.
- 1961: Spacewar!
 - MIT students Martin Graetz, Steve Russell, and Wayne Wiitanen created the game Spacewar! on a DEC PDP-1 mini-computer which also used a vector display system.



Image sources: <https://en.wikipedia.org/wiki/Spacewar!>

1966: The Brown Box



- 1966: *The Brown Box*
 - Ralph Baer, an electronics engineer working at Sanders Associates, thought again about an idea he had in 1951 when he was asked to build a television set: the possibility of using a television set to do something else than watch broadcasts. Several other prototypes were built between 1966 and 1968, the most complete being *The Brown Box*.



Image sources: <https://www.polygon.com/2014/12/7/7350497/ralph-baer-obituary>

1972: Magnavox Odyssey



- Ralph Baer signed an agreement with Magnavox and Odyssey to produce a game system based on his Brown Box prototype in 1971.
- The first home video game system!



Image sources: <http://www.warpedfactor.com/2015/08/video-game-firsts-magnavox-odyssey.html>

1971 - 1974: Rise of the Arcades



- 1971: Computer Space
 - Created by Nolan Bushnell who was inspired by Spacewar!
 - The first arcade game with a coin slot and hence the birth of the video game industry!
- 1972: PONG
 - Atari's PONG became the first hit video game.



Computer
Space



PONG / Nolan
Bushnell

Image sources: <http://www.skooldays.com/categories/arcade/ag1028.htm>, <https://www.videogamesblogger.com/2009/02/06/atari-founder-and-pong-creator-nolan-bushnell-to-receive-british-academy-video-games-award.htm>

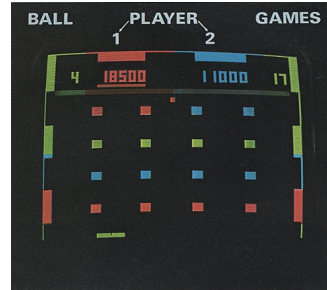
1971 - 1974: Rise of the Arcades



- Some popular arcades of the early 70s



Gotcha



TV Pin Game



Space Race

Image sources: [https://en.wikipedia.org/wiki/Gotcha_\(video_game\)](https://en.wikipedia.org/wiki/Gotcha_(video_game)), https://www.arcade-museum.com/game_detail.php?game_id=10001, <https://flyers.arcade-museum.com/?page=flyer&db=videodb&id=1255&image=1>, https://www.arcade-museum.com/game_detail.php?game_id=9681, http://www.old-computers.com/museum/software_detail.asp?id=464

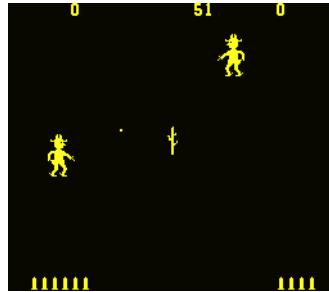
1974 - 1975: Technical innovations



- Tank is first game to store graphics data on a ROM chip.
- Gun Fight is the first game to use a microprocessor.
- Steeplechase is the first six-player arcade video game.
- Indy 800 is the first eight-player game, complete with a steering wheel and foot pedals for each of eight players (it also used a color CRT).



Tank



Gun Fight



Steeplechase



Indy 800

Image sources: <https://japeland.wordpress.com/2013/03/28/5050-video-games-10-11-tanks-for-the-memories/>, <https://www.arcade-history.com/?n=gun-fight-model-597&page=detail&id=1040>, <https://flyers.arcade-museum.com/?page=flyer&db=videodb&id=3586&image=2>, <http://cmuems.com/2015b/category/lectures/>

1976: More innovations



- General Instruments's AY-3-8500 chip is released.
 - It had all of the circuitry necessary for a video game on a single chip, making the production of video games much easier.
- The Fairchild/Zircon Channel F, the first cartridge-based home game system, is released.
 - Before this, home game consoles had their games hard-wired into their system.
- Night Driver is the first game to simulate a first-person perspective.



Telstar Classic
(based on the AY-3-8500)



Fairchild/Zircon Channel F



Night Driver

Image sources: <http://www.computercloset.org/ColecoTelstarClassic.htm>, <http://atariage.com/forums/topic/147468-fairchild-channel-f/page-2>, <https://sprittd.com/en/news/racing-games-from-the-classic-arcade-games-to-the-realist-simulation-games>

1977: Atari / Nintendo



- Atari's VCS home console system (later renamed the 2600) is released.
 - The 2600 was the most successful video game console of its time.
- Nintendo releases its first home video game, the Color TV-Game 6.



*Atari Video Computer System
(Atari 2600)*



Nintendo Color TV-Game 6

Image sources: <https://www.engadget.com/2007/10/01/happy-30th-birthday-atari-2600/>, [https://commons.wikimedia.org/wiki/File:Nintendo_Color_TV_game_6_\(Cut_out\).jpg](https://commons.wikimedia.org/wiki/File:Nintendo_Color_TV_game_6_(Cut_out).jpg)

Short survey: Nintendo



- Does anybody know, what Nintendo has done before 1977?
 - Founded in 1889 by Fusajirō Yamauchi in Kyōto
 - Leading producer of Hanafuda-Cards



Image sources: <https://www.wired.com/2010/09/0923nintendo-founded/>

Short survey: Console generations



- Consoles are often categorized in generations.
 - 1977 - 1983: Atari 2600 was the dominant console in the 2nd generation.
 - Currently we are in the 8th/9th generation.

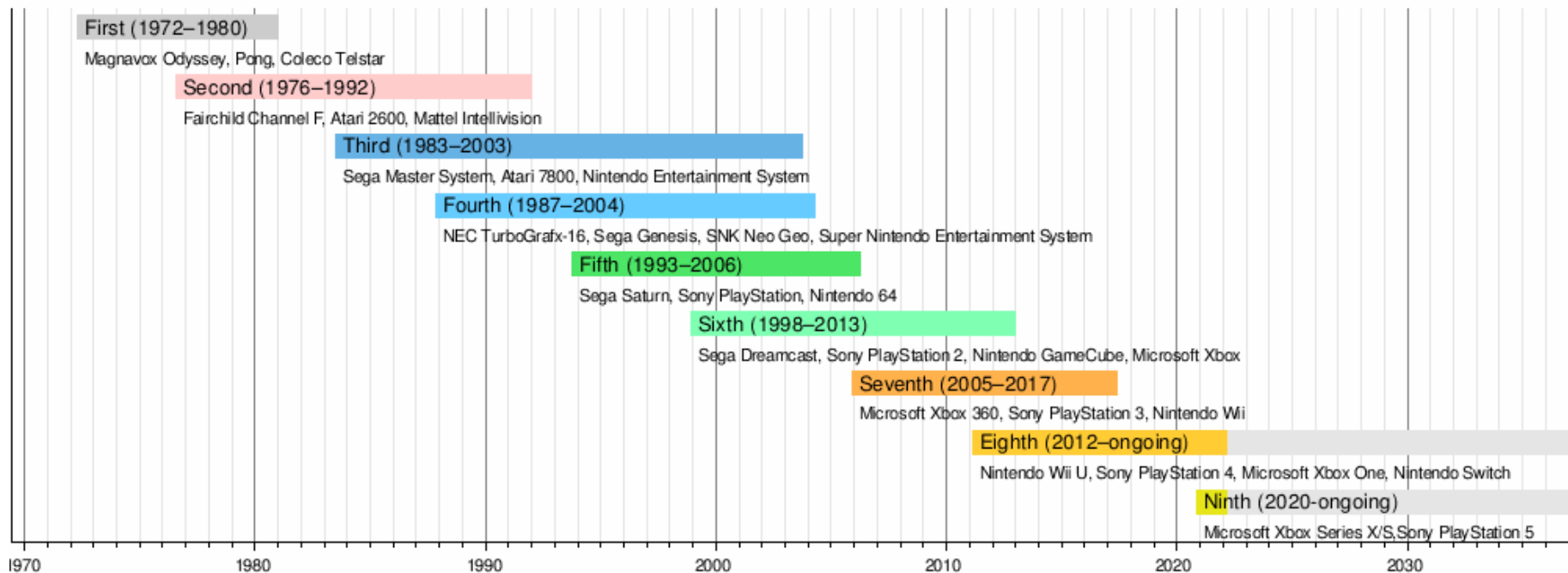


Image sources: <https://upload.wikimedia.org/wikipedia/en/timeline/jy4ton1yrfxofnr4irhchlufuqrlv3pg.png>

Some 2nd-gen consoles



Coleco Colecovision



Magnavox Odyssey 2



Mattel Intellivision



*Intellivoice
(Intellivision Add-on)*

Image sources: <http://www.theoldcomputer.com/roms/index.php?folder=Coleco/ColecoVision>, <http://thevideogamesystems.blogspot.de/2011/04/magnavox-odyssey-2-game-system-that.html>, <https://en.wikipedia.org/wiki/Intellivision>

1978 - 1979: Popular arcade games



- 1978: Space Invaders
 - Inspiration for many vertical shooting games to follow.
- 1979: Warrior
 - First one-on-one fighting game.
- 1979: Asteroids and Lunar Lander
 - Both vector graphics games.
- 1979: Galaxian
 - The first game to have 100 percent of its graphics in RGB color.



Space Invaders



Warrior



Asteroids



Galaxian

Image sources: <https://www.pinterest.com/pin/209206345165840467/>, <http://www.andysarcade.de/warrior.html>, <https://www.usgamer.net/articles/the-day-i-melted-an-asteroids-machine>, <https://en.wikipedia.org/wiki/Galaxian>

1979 - 1980: Popular arcade games



- 1979: Puck-Man is released
- 1980: Renamed to Pac-Man for its North American release.
- 1980: Battlezone and Defender
 - Battlezone is the first arcade game to feature a true 3D environment.
- 1980: Ultima
 - The first home computer game with 4-directional scrolling.
- 1980: Star Fire
 - The first cockpit game, and the first arcade game to feature a high-score table using players' initials.



Puck-Man



Battlezone



Ultima



Star Fire

Image sources: <http://www.imdb.com/title/tt0260264/>, <http://www.ocdgame.dk/game/?id=566>, https://en.wikipedia.org/wiki/Ultima_I:_The_First_Age_of_Darkness, https://en.wikipedia.org/wiki/Star_Fire

1981 - 1982: Popular arcade games



- 1981: Donkey Kong
 - The United States arcade game industry reaches \$5 billion.
- 1982: Q*Bert
 - One of the most merchandised arcade games behind Pac-Man and Donkey Kong
- 1982: Zaxxon
 - The first arcade game to be advertised on television.
 - <http://www.youtube.com/watch?v=Z0bxTjxqwxg>



Donkey Kong



*Q*Bert*



Zaxxon

Image sources: [https://en.wikipedia.org/wiki/Donkey_Kong_\(video_game\)](https://en.wikipedia.org/wiki/Donkey_Kong_(video_game)), https://en.wikipedia.org/wiki/Q*bert, <https://en.wikipedia.org/wiki/Zaxxon>

1983: Dragon's Lair



- The first analog laserdisc-based coin-op video game to be released.
- It was so impressive that it cost 50 cents instead of 25 cents to play, and some arcade owners even installed a monitor over the cabinet so people could watch the game as it was played.
- Created the genre of Interactive Movies.
- However, the gameplay was very limited.
 - Pressing the right buttons at the right time.
 - This style of gameplay is referred to as QTEs (Quick Time Events) today.
- Featured in Netflix's Stranger Things in 2017.



Image sources: <http://www.replaymuseum.org/games/dragons-lair/>, <https://courtneyhilbig.com/2017/05/24/dragonslair/>

1983: I, Robot



- The first raster video game with filled-polygon three-dimensional graphics and flat shading
- Also the first video game to feature camera control options.
- Upon release, it received poor reception and was a financial flop.
 - Why was it a flop?

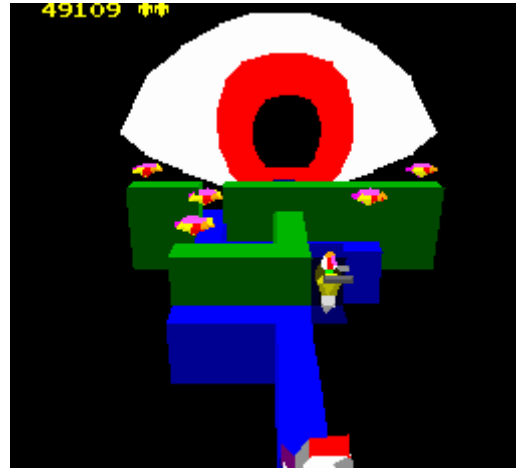


Image sources: <http://www.hardcoregaming101.net/i-robot/>, <https://www.classicarcadecabinets.com/i-robot.html>

The big crash of 1983

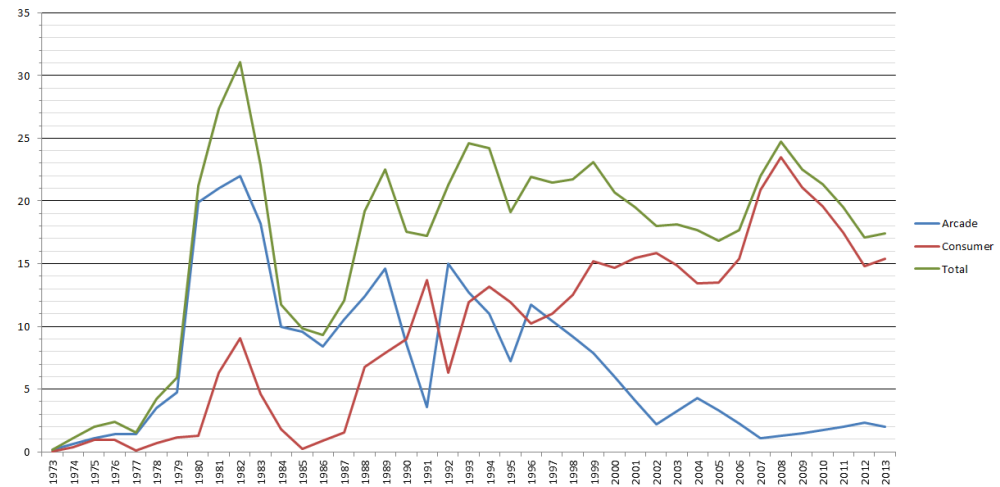


- The games industry (at least in the USA) severely crashed.
 - Many people blamed the Atari game ET for this...
 - But it was more due to an oversaturated market combined with poor quality control.
 - Could this happen again?
 - The mobile game market now looks very similar to what was seen in 1983!
- <https://www.youtube.com/watch?v=ikleaCE3Ljc>



Image sources: http://vgsales.wikia.com/wiki/Video_game_industry

U.S. Video Game Market Revenues
(Billion US Dollars, inflation adjusted for 2012)



1984 - 1995: 3rd gen (8-bit era)



- 1984: Nintendo releases the Famicom system in Japan.
- 1985: Nintendo releases the NES (Nintendo Entertainment System)
 - A new version of the Famicom.
 - Its popularity helped to bring an end to the industry crash.
 - Nintendo also releases Super Mario Bros., which becomes one of the best-selling games of all time.



Famicom



NES

Image sources: <https://arstechnica.com/gaming/2013/07/time-to-feel-old-inside-the-nes-on-its-30th-birthday/>, <http://juniordsports.com/top-10-sports-games-on-the-nintendo-entertainment-system/>

1985: The famous plumber



- Shigeru Miyamoto
 - He reworked the character Jumpman from Donkey Kong into Mario, and gave him a brother: Luigi
 - 1983: The first Mario title was the arcade game Mario Bros.
 - 1985: Release of Super Mario Bros. for the NES
 - One of the best selling games of all time!
- Koji Kondo
 - Composer of the Super Mario Bros. sound track
 - The musical elements were adjusted to the expected reactions of the player.



Image sources: https://en.wikipedia.org/wiki/Mario_Bros., https://en.wikipedia.org/wiki/Super_Mario_Bros., http://mariokart.wikia.com/wiki/Shigeru_Miyamoto, <https://www.pinterest.com/pin/450078556492716862/>

1985: Four addictive blocks



- Alex Pajitnov designs Tetris
 - The name is derived from the Greek numerical prefix tetra- (all of the game's pieces contain four segments) and tennis.
 - 1986: The game rights were first sold to Mirrorsoft (UK) and Spectrum Holobyte.
 - 1988: The Soviet government began to market the rights to Tetris via an organization named Elektronorgtechnica and within a few years the game turned into a massive success.
 - Pajitnov did not get much money from the deal, even though, for example, Nintendo got an advantageous return from selling the game.
 - Electronic Gaming Monthly's 100th issue had Tetris in first place as "Greatest Game of All Time".
 - Also one of the best selling games of all time.

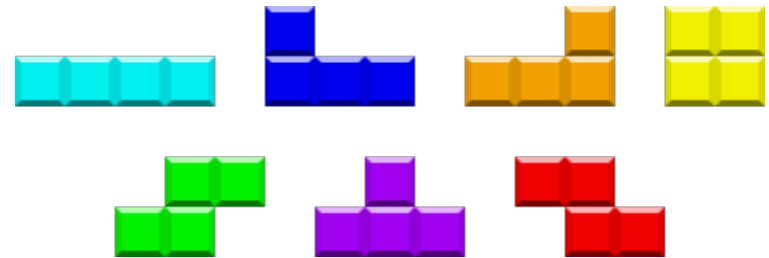


Image sources: https://commons.wikimedia.org/wiki/File:Tetrominoes_IJLO_STZ_Worlds.svg

1986: The boy in green



- 1986: The Legend of Zelda for the NES
 - Invented by Shigeru Miyamoto and Takashi Tezuka.
 - The game features a mixture of action, puzzles, adventure/battle gameplay, exploration, and questing.
 - Nintendo's MMC1 chip allowed the player to save his progress (battery pack).
 - One of Nintendo's most popular franchises, selling millions of copies worldwide.
 - The Legend of Zelda: Ocarina of Time (N64) holds the record for the highest reviewed video game of all time with a Metacritic score of 99/100.



Image sources: https://thewellredimage.com/2016/03/19/the-legend-of-zelda-first-quest/evolution_of_in_game_link_by_zerjer97-d630e05/, http://vsbattles.wikia.com/wiki/File:Link_breath_of_the_wild_render_by_jrren_ders-dad872p.png

1986: Sega enters the competition



- 1986: Release of the Sega Master System
 - Very similar to the NES. Both consoles came with similar-looking controller pads and both were sold with light guns for target shooting games.
- Although many in the industry declared that the SMS technology was superior to the NES, the buying public favoured the NES.



Master System



Master System II

Image sources: <https://www.warpzoneonline.com/product/sega-master-system/>, http://gaming.wikia.com/wiki/Master_System_II

1987 - 1999: 4th gen (16-bit era)



- 1987: Hudson Soft and NEC release the PC Engine (TurboGrafx-16)
 - First console with an optional CD add-on.
- 1988: Sega Mega Drive
 - <https://www.youtube.com/watch?v=65E53rNC1io>
- 1990: Super Nintendo Entertainment System
 - Despite stiff competition from Sega's Genesis/Mega Drive console, the Super NES eventually took the top selling position, selling 49.10 million units worldwide.
- 1990: SNK releases the Neo Geo (“home arcade platform”)



TurboGrafx-16



Sega Genesis



SNES



Neo Geo

Image sources: <https://www.theverge.com/2016/2/27/11125294/kanye-west-new-album-turbografx-16>, https://en.wikipedia.org/wiki/Sega_Genesis, <https://commons.wikimedia.org/wiki/File:SNES-USA.png>, http://www.bitc.com/en_gb/2014/02/04/200-a-game-we-look-at-the-90s-powerhouse-console-the-neo-geo/

1987: Maniac Mansion



- The first adventure game with a point-and-click interface.
- Introduction of the SCUMM (Script Creation Utility for Maniac Mansion) engine.
 - Written by Ron Gilbert and Aric Wilmunder of Lucasfilm Games (now LucasArts)
- Lucasfilm Games would go on to use the SCUMM engine (and updated versions of it) for their games into the 1990s.



Image sources: <http://hol.abime.net/2405>

First CD-ROM Game



- 1987: Cyan's The Manhole was the first computer game distributed on CD-ROM
- Children's games can also be innovative, even in terms of technology!
- Pros of CD-ROM:
 - Large capacity
 - High fidelity audio (music, speech)
 - Full Motion Video
 - Production is very cheap
- Cons of CD-ROM:
 - Long load times
 - Piracy is much easier (when CD writers became available)
 - Need memory cards or HDDs to save progress
 - CDs break very easily (dust, scratches, age, ...)
 - Cartridges could contain additional HW (E.g. the SNES's FX Chip)

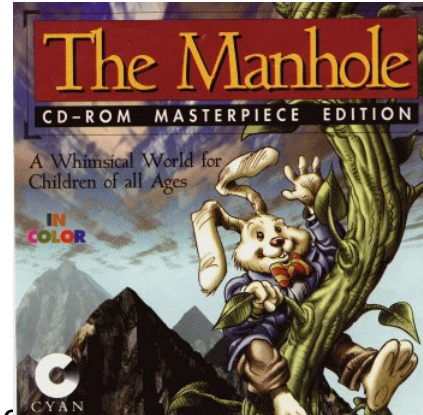


Image sources: <http://www.adventureclassicgaming.com/index.php/gallery/category/C601/>

1989 - 1990: First handheld consoles



- 1989: Nintendo Game Boy
 - The Game Boy and Game Boy Color combined have sold 118.69 million units worldwide.
- 1989: Atari Lynx
 - The Lynx enjoyed a small success. It was released shortly after Nintendo's Game Boy, which, despite its monochrome screen, proved more popular thanks to its lower price and its being packaged with Tetris.
- 1990: Sega Game Gear
 - Despite costing more than the Lynx, being larger in size, and having a lower battery life, the Game Gear outsold the Lynx.



Game Boy



Lynx



Game Gear

Image sources: https://de.wikipedia.org/wiki/Game_Boy, https://en.wikipedia.org/wiki/Atari_Lynx, http://smurfs.wikia.com/wiki/Game_Gear

1992: Employment of the FPS genre



- 1992: id Software releases Wolfenstein 3D
 - Was released as shareware, which allowed it to be copied widely.
 - The game is widely regarded by critics and game journalists as having established the basic run-and-gun archetype for subsequent FPS games.
 - John D. Carmack was Lead Programmer
 - Inspired by a technology demo of Looking Glass Studios' and the first-person role-playing video game, Ultima Underworld The Stygian Abyss from 1991. Carmack claimed he could make a faster renderer. In this he was successful.
- 1993: id Software releases Doom



Wolfenstein 3D



*Ultima Underworld:
The Stygian Abyss*



John D. Carmack

Image sources: https://en.wikipedia.org/wiki/Wolfenstein_3D, https://en.wikipedia.org/wiki/Ultima_Underworld:_The_Stygian_Abyss, <https://kotaku.com/memories-of-doom-by-john-romero-john-carmack-1480437464>

1993 - 2006: 5th gen (32 and 64 bit)



- 1993: Atari Jaguar
- 1993: 3DO Interactive Multiplayer
- 1994: Sega Saturn
- 1994: PC-FX
- 1994: Sony PlayStation
 - The PlayStation quickly outsold all of its competitors, with the exception of the aging Super Nintendo Entertainment System, which still had the support of many major game companies.
- 1996: Nintendo 64
- The fifth generation is most noted for the rise of fully 3D games.
- A slow decline of cartridges in favour of CDs, which allowed much greater storage capacity than what was previously possible.



Jaguar



Saturn



PlayStation



N64

Image sources: http://crappy-games.wikia.com/wiki/Atari_Jaguar, https://en.wikipedia.org/wiki/Sega_Saturn, http://playstationhistory.wikia.com/wiki/Playstation_1, http://de.nintendo.wikia.com/wiki/Nintendo_64

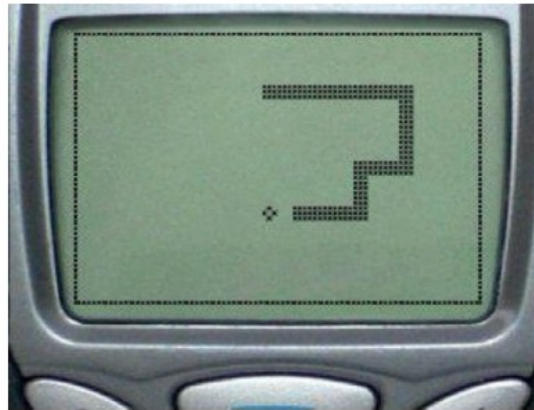
1994: First Cell Phone Games



- A mobile game is a video game played on a mobile phone, smartphone, PDA, tablet computer, portable media player or calculator.
- 1994: Tetris on the Hagenuk MT-2000 device.
- 1997: Nokia launches the very successful Snake.
- Today, cell phone games have come a very long way. Their graphics are about the same as you would expect from 1 or 2 console generations ago.



MT-2000



Snake



Fortnite mobile

Image sources: https://en.wikipedia.org/wiki/Hagenuk_MT-2000, <http://www.newstalk.com/5-things-you-didnt-know-about-Snake>, <https://www.polygon.com/2018/3/15/17126798/fortnite-battle-royale-mobile-invite-ios-epic-games-app-store-first-wave>

1996: The 3D Revolution



- 1996: 3dfx Interactive releases the first hardware accelerated 3D graphics card: The 3dfx Voodoo.
 - It only rendered 3D; it had to be piggybacked with a 2D video card.
 - Programmed using the Glide 3D graphics API.
 - Brought arcade graphics into the home.



Image sources: <http://www.pcgameshardware.de/Grafikkarten-Grafikkarte-97980/News/3dfx-stellt-die-Voodoo-Graphics-vor-PCGH-Retro-Spezial-698841/>, https://en.wikipedia.org/wiki/3dfx_Interactive, <http://www.dnsdata.com/M3D2.htm>

1996: Tomb Raider



- Eidos released Tomb Raider in 1996 which introduced the female adventurer Lara Croft
 - Lara was created by designer Toby Gard, who admitted in an interview that her large breasts occurred by accident when his mouse slipped...
 - Tomb Raider games have sold over 45 million copies worldwide, becoming one of the best-selling video game franchises.
 - Lara Croft herself has become one of the most recognizable video game protagonists in existence.



Image sources: <http://www.gamesgrabr.com/blog/2016/01/07/the-evolution-of-lara-croft/>

1998 – 2004: 6th generation



- 1998: Sega Dreamcast
- 2000: Sony PlayStation 2
 - The PlayStation 2 is the best-selling console of all time, having reached over 154.4 million units sold as of November 21, 2011.
- 2001: Nintendo GameCube
- 2001: Microsoft Xbox
 - 2002: Xbox Live online gaming service begins.



Dreamcast



PlayStation 2



GameCube



XBox

Image sources: <https://levelskip.com/consoles/Sega-Dreamcast-still-best-fighting-game-console>, https://en.wikipedia.org/wiki/PlayStation_2, http://animalcrossing.wikia.com/wiki/Nintendo_GameCube, [https://en.wikipedia.org/wiki/Xbox_\(console\)](https://en.wikipedia.org/wiki/Xbox_(console))

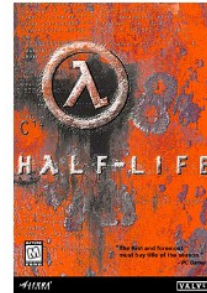
1998: Game Design innovations



- 1998: Konami releases Dance Dance Revolution and the first games in its Beatmania series and GuitarFreaks series.
 - Predecessors of the popular Guitar Hero and Rock Band series.
- 1998: Sierra Studios releases Half-Life.
 - Its gameplay influenced the design of first-person shooters for years after its release, and it is widely considered to be one of the greatest computer games of all time
- 1998: Rockstar Games releases Grand Theft Auto.
 - The GTA series had major cultural impact and is one of the most popular representatives of the open-world genre.



GuitarFreaks



Half-Life



Grand Theft Auto

Image sources: https://www.arcade-museum.com/game_detail.php?game_id=8027, [https://en.wikipedia.org/wiki/Half-Life_\(video_game\)](https://en.wikipedia.org/wiki/Half-Life_(video_game)), [https://en.wikipedia.org/wiki/Grand_Theft_Auto_\(video_game\)](https://en.wikipedia.org/wiki/Grand_Theft_Auto_(video_game))

2000: Simulated life



- 2000: Maxis's The Sims is released.
 - Will Wright's The Sims was a revolutionary product that required the creator to engage in significant research as well as to convince the team and publisher to come on board the project.
 - Critics praised it with positive reviews. It has been a success in many ways - attracting casual gamers and female gamers (the latter making up ~50% of gamers today).
- 2002: The Sims overtakes Myst and becomes the best-selling home computer game at the time.

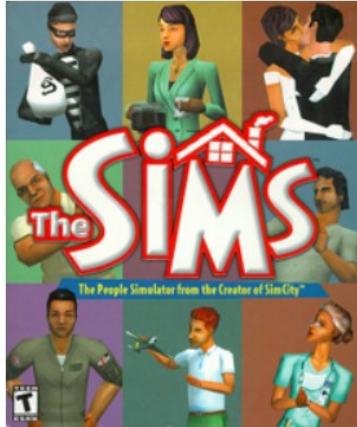


Image sources: [https://en.wikipedia.org/wiki/The_Sims_\(video_game\)](https://en.wikipedia.org/wiki/The_Sims_(video_game))

2004 – 2013: 7th generation



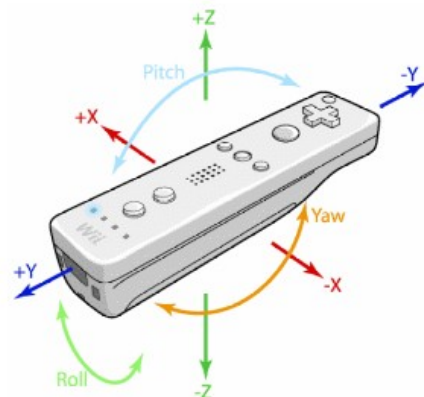
- 2004: Nintendo DS
- 2004: Sony PlayStation Portable
- 2005: Microsoft Xbox 360
- 2006: Sony PlayStation 3
- 2006: Nintendo Wii
 - Technologically inferior, but still the most successful!
- Large increases in development budgets
- Nintendo capitalizes on casual gaming
- Motion controls revolutionize gameplay
 - Playable tutorials became more important than ever before!
 - Games are now more accessible.
 - Children, seniors, handicapped people, etc.



PlayStation Move



Xbox 360 Kinect



Nintendo Wiimote

Image sources: https://en.wikipedia.org/wiki/PlayStation_Move, <http://kinect.wikia.com/wiki/File:Kinect.jpg>, <https://leicesterrasberrypi.wordpress.com/physics-with-a-wiimote/>

2011 – present: 8th generation



- 2011: Nintendo 3DS
- 2011: Sony PlayStation Vita
- 2012: Nintendo Wii U
- 2013: Sony PlayStation 4
 - 2016: Playstation 4 Pro
- 2013: Microsoft Xbox One
 - 2017: Xbox One X
- Connectivity and collaborative gaming seems to be the most important aspect of the 8th generation.



3DS



PSVita



Wii U



PlayStation 4



Xbox One

Image sources: https://en.wikipedia.org/wiki/Nintendo_3DS, http://finalfantasy.wikia.com/wiki/PlayStation_Vita, http://animalcrossing.wikia.com/wiki/Wii_U, http://gamegrumps.wikia.com/wiki/PlayStation_4, http://finalfantasy.wikia.com/wiki/Xbox_One

Recent developments



- Consumer VR Devices
- Nintendo Switch console (“9th gen”, sort of...)
 - Home / portable hybrid
 - Based on NVIDIA's Tegra X1 chip
 - 2025: “Switch 2”
- 9th Console generation in 2020/2021
 - PlayStation 5
 - “Pro” version will launch in Nov. 2024
 - Xbox Series S/X
- Real time ray tracing gets widely adopted
- Unreal 5 promises “super-realistic graphics”
 - Many industry giants adopt unreal technology
- Valve releases the Steam Deck in 2022 with decent success
 - **Thanks to Proton, Linux gaming has become a viable option!**



Image sources: <https://www.techradar.com/news/world-of-tech/future-tech/the-vr-race-who-s-closest-to-making-vr-a-reality-1266538>, https://en.wikipedia.org/wiki/Nintendo_Switch, <https://www.imgtec.com/blog/unreal-engine-and-the-ray-tracing-revelation/>, <https://www.tomsguide.com/features/ps5-and-xbox-series-x-ray-tracing-heres-why-its-a-big-deal>, [https://image.goxip.com/oHJFG0SQg6BB4r1J44Tsb7VS0ug=/fit-in/500x500/filters:format\(jpg\):quality\(80\):fill\(white\)/https:%2F%2Fimages.stockx.com%2Fimages%2FValves-Steam-Deck-EU-Plug-64-GB.jpg](https://image.goxip.com/oHJFG0SQg6BB4r1J44Tsb7VS0ug=/fit-in/500x500/filters:format(jpg):quality(80):fill(white)/https:%2F%2Fimages.stockx.com%2Fimages%2FValves-Steam-Deck-EU-Plug-64-GB.jpg)

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